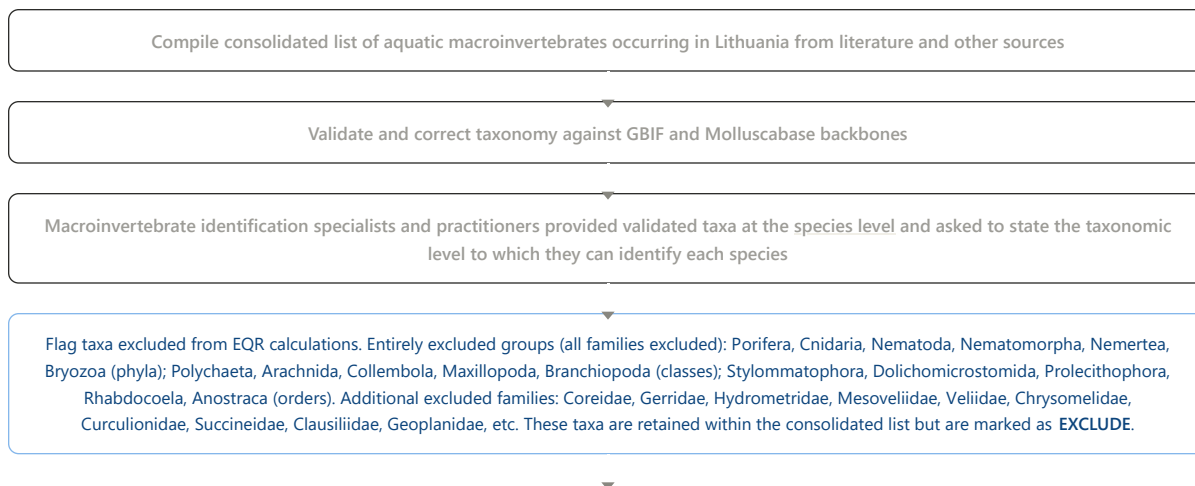


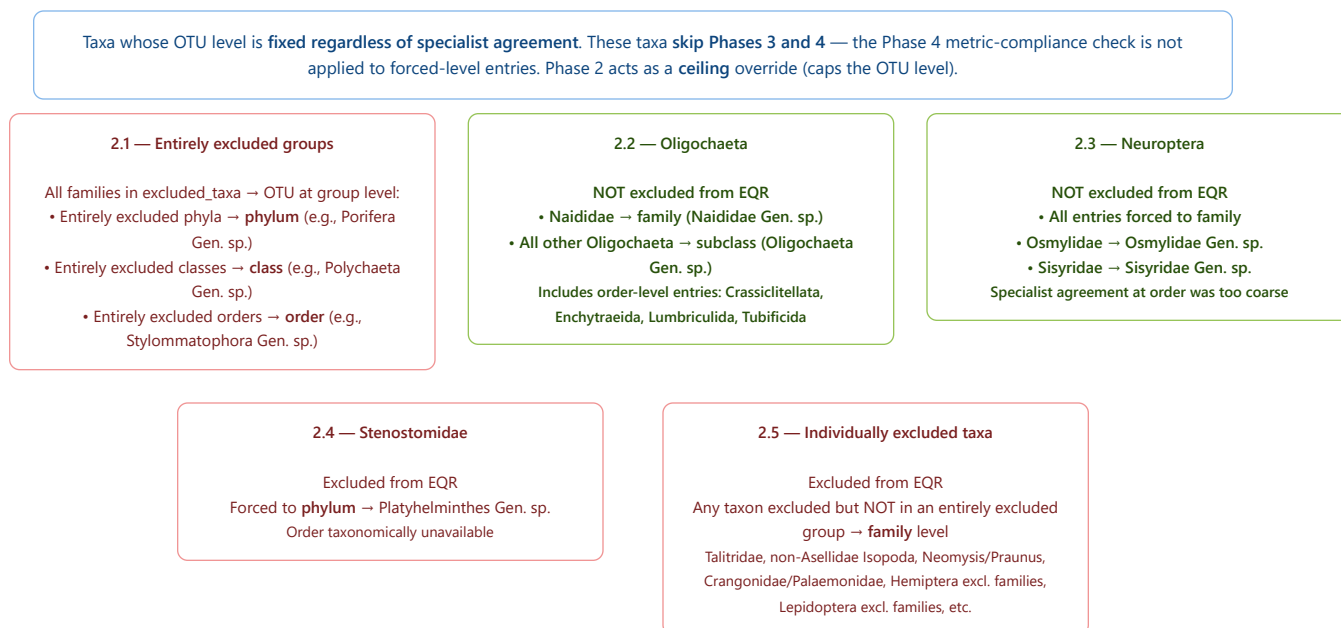
Decision tree used for assigning operational taxonomic units (OTUs)

☐ Preparatory step ☐ Process / action ☐ Decision ☐ OTU assignment ☐ Metric compliance override ☐ Annotation ☒ Minimum metric requirement

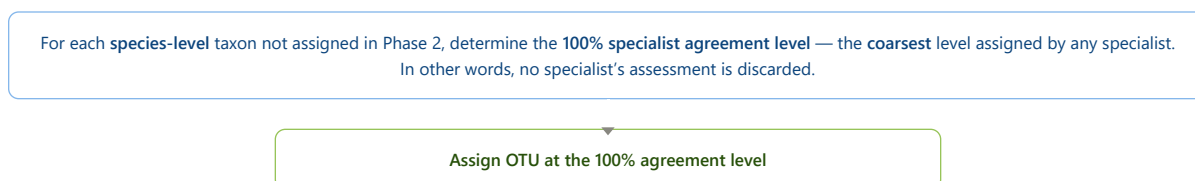
Phase 1 — Preparation



Phase 2 — Forced-level assignments



Phase 3 — 100% specialist agreement (species-level entries only)



Higher-level entries — genus, family, and above

Family, order, subclass, class, and phylum entries are structural fallback entries. They accommodate damaged, juvenile, or poorly preserved specimens and are always assigned at their own taxonomic level.

Family → family
e.g., Baetidae Gen. sp.

Order → order
e.g., Ephemeroptera Gen. sp.

Subclass / class / phylum → respective level

Truly aquatic rule (order+ entries): Only truly aquatic groups keep order-and-above entries for EQR calculations: Bivalvia (class), Plecoptera, Ephemeroptera, Odonata, Megaloptera, Trichoptera (orders). All other groups' order+ entries are excluded from EQR.

Genus entries are metric-minimum-aware, capped at genus. The metric minimum is determined by the genus's own taxonomy (group, family), NOT inherited from species-level BMWP exceptions:

- If the genus has a **forced-level override** (Phase 2) → resolves to the forced level (e.g., Talitridae genus → Talitridae Gen. sp.)
 - If the minimum is **genus or finer** → genus entry stays at **genus** level (e.g., *Leuctra* sp. — Plecoptera min = genus)
- If the minimum is **coarser than genus** → genus entry resolves to that coarser level (e.g., *Baetis* → Baetidae Gen. sp. — Ephemeroptera min = family)
 - If **no metric group** → genus entry defaults to **genus** level
 - Genus entries in **entirely excluded groups** collapse to the group-level OTU

Subspecies entries (e.g., *Nemoura cinerea* subsp. *cinerea*) that are not present in the Specialist taxalist **inherit their OTU assignment from the parent species**. This prevents subspecies from collapsing to a coarser fallback level when the parent species has a known specialist agreement.

Phase 4 — Metric compliance check (species-level entries only, applied after Phase 3)

For each species-level taxon, verify that the OTU level from Phase 3 meets or exceeds the minimum identification requirement for its group (based on the LRMI reference table). If the taxon's group has no specific metric requirement, this check is automatically passed.

Does the assigned OTU level meet or exceed the minimum metric requirement?

YES

OTU assignment is confirmed.

Final output

Final OTL entry for each taxon:

Validated name · OTU level · OTU name (e.g., *Baetis* sp. / Baetidae Gen. sp.) · Taxonomic rank · DSFI and index exclusion flags (if applicable)

Hierarchical entries (both fine-resolution OTUs and coarser fallback entries) coexist in the taxalist, accommodating damaged, juvenile, or poorly preserved specimens.

Reference table — Minimum metric requirements by group

Group	DSFI requirement	BMWP/ASPT (all scored taxa)	Other metrics	Resulting minimum
Malacostraca	<i>Asellus</i> , <i>Gammarus</i> at genus (IG + diversity)	Score 8: Astacidae [f], Score 6: Gammaridae [f], Corophiidae [f], Crangonyctidae [f], Niphargidae [f], Pontogammaridae [f]. Score 3: Asellidae [f]. Excluded taxa (Talitridae, non-Asellidae Isopoda, Neomysis/Praunus, Crangonidae/Palaemonidae) → family level, excl. EQR.	%CrHi pools Malacostraca	Genus for <i>Asellus</i>, <i>Gammarus</i>; family for others
Gastropoda	<i>Ancylus</i> , <i>Lymnaea</i> at genus (diversity)	Score 6: Viviparidae [f], Neritidae [f], Ancyliidae [f], <i>Ancylus</i> [g], Acroloxidae [f]. Score 5: Hydrobiidae [f]. Score 3: Lymnaeidae [f], Physidae [f], Planorbidae [f], Valvatidae [f], Bithyniidae [f].	—	Genus for <i>Ancylus</i>, <i>Lymnaea</i>; family for others
Bivalvia	<i>Sphaerium</i> at genus (diversity)	Score 6: Unionidae [f]. Score 3: Sphaeriidae [f].	—	Genus for <i>Sphaerium</i>; family for others

Plecoptera	13 genera at genus (IG1/IG2). Families: Leuctridae, Capniidae, Perlodidae, Perlidae, Chloroperlidae (IG1); Taeniopterygidae, Nemouridae (IG2).	Score 10: Taeniopterygidae [f], Leuctridae [f], Perlodidae [f], Perlidae [f], Chloroperlidae [f], Capniidae [f]. Score 7: Nemouridae [f].	#DEP: species	At least genus
Ephemeroptera	8 families: Ephemeridae (IG1), Ametropodidae, Ephemerellidae, Heptageniidae, Leptophlebiidae, Siphonuridae (IG2), Caenidae (IG3), Baetidae (IG5).	Score 10: Siphonuridae [f], Heptageniidae [f], Leptophlebiidae [f], Ephemerellidae [f], Potamanthidae [f], Ephemeridae [f]. Score 7: Caenidae [f]. Score 4: Baetidae [f].	#DEP: species	At least family
Trichoptera	<u>Case-bearing</u> : Beraeidae, Brachycentridae, Goeridae, Glossosomatidae, Hydroptilidae, Leptoceridae, Lepidostomatidae, Limnephilidae, Molannidae, Odontoceridae, Phryganeidae, Sericostomatidae. <u>Caseless</u> : Ecnomidae, Hydropsychidae, Philopotamidae, Polycentropodidae, Psychomyiidae, Rhyacophilidae.	Score 10: Beraeidae [f], Brachycentridae [f], Goeridae [f], Lepidostomatidae [f], Leptoceridae [f], Molannidae [f], Odontoceridae [f], Phryganeidae [f], Sericostomatidae [f]. Score 8: Ecnomidae [f], Philopotamidae [f], Psychomyiidae [f]. Score 7: Apataniidae [f], Glossosomatidae [f], Limnephilidae [f], Polycentropodidae [f], Rhyacophilidae [f]. Score 6: Hydroptilidae [f]. Score 5: Hydropsychidae [f].	—	At least family
Coleoptera	<i>Elmis</i> , <i>Limnius</i> , <i>Elodes</i> at genus (IG1/IG2 + diversity)	Score 5: Haliplidae [f], Hygrobiidae [f], Noteridae [f], Dytiscidae [f], Gyrinidae [f], Hydrophilidae [f], Hydraenidae [f], Hydrochidae [f], Helophoridae [f], Georissidae [f], Clambidae [f], Scirtidae [f], Helodidae [f], Dryopidae [f], Elmidae [f], Chrysomelidae [f], Curculionidae [f].	—	Genus for <i>Elmis</i> , <i>Limnius</i> , <i>Elodes</i> ; family for others
Diptera	<i>Chironomus</i> at genus ; Eristalinae at subfamily ; Simuliidae, Psychodidae, Chironomidae at family	Score 5: Tipulidae [f], Simuliidae [f], Limoniidae [f], Pediciidae [f], Cylindrotomidae [f]. Score 2: Chironomidae [f].	#DEP: family	Family ; genus for <i>Chironomus</i> ; subfamily for Eristalinae
Megaloptera	<i>Sialis</i> at genus (IG4 + neg. diversity)	Score 4: Sialidae [f].	—	Genus for <i>Sialis</i>
Hirudinea	<i>Erpobdella</i> , <i>Helobdella</i> at genus (neg. diversity)	Score 4: Piscicolidae [f]. Score 3: Glossiphoniidae [f], Hirudinidae [f], Erpobdellidae [f].	%CrHi pools Hirudinea	Genus for <i>Erpobdella</i> , <i>Helobdella</i> ; family for others
Oligochaeta	Naididae at family (IG6; formerly Tubificidae); Oligochaeta ≥100 individuals as precluser	Score 1: Oligochaeta [sc], Naididae [f].	—	Naididae at family ; others at subclass
Hemiptera	—	Score 10: Aphelocheiridae [f]. Score 5: Nepidae [f], Naucoridae [f], Notonectidae [f], Pleidae [f], Corixidae [f], Gerridae [f], Hydrometridae [f], Mesoveliidae [f].	%EHP pools Hemiptera. Excluded families → family-level OTU.	At least family
Odonata	—	Score 8: Lestidae [f], Calopterygidae [f], Gomphidae [f], Cordulegastridae [f], Aeshnidae [f], Corduliidae [f], Libellulidae [f]. Score 6: Coenagrionidae [f], Platycnemididae [f].	—	At least family
Turbellaria	Tricladida at order (pos. diversity)	Score 5: Planariidae [f], Dendrocoelidae [f], Dugesiidae [f].	—	At least family
Lepidoptera	—	—	—	At least family
Neuroptera	—	—	&mdash.	At least family